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A novel inclined membrane contact model for analyzing the pneu-net soft actuator lateral wall contact

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Abstract

Pneu-net soft actuators are widely used in the soft robotics society owing to their light weight, high deformation, and fast response. This paper presents a novel theoretical framework to model the static analysis and contact mechanics of pneumatic soft actuators undergoing large deformations. While most soft robots exhibit complex material behaviors, we show that their mechanics can be accurately captured through the fundamental principles of elasticity and contact theories. The core contribution is an inclined membrane contact model that elegantly transforms the complex three-dimensional contact between angled surfaces into an equivalent problem of horizontal contact, enabling the use of established contact solutions. This model is integrated with an energy-based solution for elastic deformation to fully characterize soft actuator bending. The generalizable modeling approach is applied to the example of a pneumatic net actuator, with comprehensive validation against finite element analysis and experiments. This work advances a fundamental understanding of soft machine statics and contact mechanics while providing an analytical tool for the design and control of deformable actuators. The flexible theoretical framework presented can be extended to diverse interdisciplinary problems involving moving surfaces in contact.

Keywords: soft pneumatic actuator, soft robot, analytical modeling, contact mechanic, membrane contact theory

1. Introduction

Soft robotic systems composed of highly deformable elastomers have gained significant interest due to their capabilities for flexibly navigating environments and performing tasks that challenge more rigid robots [1]. However, the analytical modeling of their continuous deformation mechanics presents considerable theoretical challenges [2]. These hyperelastic materials undergo virtually infinite degrees of freedom with nonlinear stress-strain behaviors [3]. This necessitates developing

modeling techniques firmly rooted in the fundamental principles of elasticity theory, contact mechanics, machine kinematics, and applied mathematics [4].

One prominent type of soft robot is the pneu-net soft actuator (PNSA) [5], which leverages pneumatic power for its operation. PNSAs are made up of chambers that are connected to each other and a bottom layer that cannot be stretched. This makes them light but very flexible, and they can respond quickly. Establishing an accurate analytical model is helpful to predict actuators' deformation and helps us design and control actuators. However, accurately modeling PNSAs presents considerable challenges due to their complex internal structure, infinite degrees of freedom, and significant

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deformation. Although the finite element method (FEM) has been widely used for designing PNSAs and predicting their deformation with accuracy [6], its application is hindered by time-consuming computations and the absence of a robust model capable of elucidating the relationship between the actuator's deformation and associated parameters.

In recent years, there has been some research on the modeling and analysis of PNSA bending due to elastomer stretching. de Payrebrune and O'Reilly [7] developed a rod-based model for pneumatic soft robots based on Euler's elastic theory, specifically tailored for nonconstant curvature configurations. Alici *et al* [8] calculated the effective elastic modulus of PNSAs and derived an analytical model for bending. Wang and Hirai [9] simplified PNSAs by approximating them as rigid line segments, and they obtained a dynamic model using the Lagrange equation. Liu *et al* [10] obtained the relationship between the strain energy of the actuator and the work done by the air based on the energy method. Gu *et al* [11] proposed an analytical model for generalized PNSAs in three-dimensional (3D) space, leveraging the principle of minimum potential energy. Hao *et al* [12] investigated the effects of design parameters and various cross-sections on PNSA bending. Zhong *et al* [13] made an analysis model that takes into account the actuator's full axial deformation and the Neo-Hookean (NH) theory of hyperelasticity. Despite their contributions, these bending models fail to account for the influence of lateral wall contact on bending and fall short of accurately predicting the deformation of PNSAs subjected to such contact.

Recognizing the significance of lateral wall contact in actuator bending, some researchers have attempted to address this factor by developing models. Lee *et al* [14] suggested a simplified analytical model of lateral wall extrusion that uses geometric approximation and the theory of pseudo-rigid bodies. However, their model considers the overlap of surfaces freely inflated by adjacent lateral walls as the contact region and does not consider the actual deformation of the lateral walls. Wang *et al* [15] devised a theoretical model for segmented pneu-net actuators with lateral wall contact based on the principle of virtual work. Nonetheless, the model oversimplifies the system and does not take into account the influence of elastomer stretching on both the lateral walls and the gap layer. Liu *et al* [16] analyzed the bending angle of PNSAs considering lateral wall contact using Hertz theory. Nevertheless, their model assumes that adjacent lateral walls are always parallel, overlooking the gradual increase in the angle between the adjacent lateral walls following inflation.

The contact between the adjacent lateral walls of the PNSA is essentially the contact between the inclined membrane and the substrate. Sachin *et al* [17, 18] investigated this contact using a contact mechanics model for large deformations and proposed a contact model between the inclined membrane and substrate. Nevertheless, the model exhibits certain limitations. Firstly, they assume that the maximum deformation of a rectangular membrane is equivalent to that of a square membrane with the same width under the same pressure. However, the deformation of the rectangular membrane is related to the

aspect ratio. Specifically, when the aspect ratio is 2, the maximum deflection of the rectangular membrane is approximately twice that of the square membrane. Secondly, they consider the eccentricity distance associated with the contact force between lateral walls to be the distance from the initial contact point to the inextensible layer. Nevertheless, given the uniform air pressure, contact force, and nearly symmetrical nature of the contact region, the eccentricity should be the distance from the center of the contact region to the inextensible layer rather than from the initial contact point to the inextensible layer.

This work aims to address these gaps through an integrated modeling framework based on the first principles of mechanics and mathematics. The primary contributions are:

1. An energy-based solution approach capturing hyperelastic chamber deformation mechanics.
2. A novel inclined membrane contact model elegantly transforming the multidimensional contact problem into equivalent planar contact solvable by established theories.
3. Combining these solutions to provide a comprehensive model relating soft actuator morphology and properties to performance.

The resulting analytical model grants fundamental insight into the synergistic interplay between design, elasticity, contact pressures, and dynamics. This work enhances the science of soft machine mechanics, contact mechanics, applied elasticity, and dynamic systems modeling. The integrated approach establishes a generalized framework applicable to diverse robotic and deformable systems across engineering disciplines.

The subsequent sections of this paper are organized as follows: section 2 introduces the structure and bending principle of PNSAs, and section 3 develops the bending model due to elastomer stretching. In section 4, the contact model for the inclined membrane and the substrate is built, and we use it to analyze the deformation caused by the contact between adjacent lateral walls of the PNSA. Section 5 verifies our theoretical model through finite element analysis (FEA) and experimental tests. Finally, we conclude with a summary of our findings in section 6.

2. Structure and bending principle of the PNSA

The PNSA is typically made of rubber and consists of a main body, an elastomer layer, and an inextensible layer (typically paper), with a series of interconnected chambers inside. Figure 1 illustrates the overall and localized structures of the PNSA, along with the symbols corresponding to the geometric parameters of the actuator. The coordinate axes X , Y , and Z correspond to the directions of length, width, and height, respectively. There are two factors that cause PNSA deformation: elastomer stretching and adjacent lateral wall extrusion. Figure 2 illustrates the finite element (FE) simulation results of the actuator deformation. When chambers are inflated, the main body is stretched, and the lateral walls expand. Since

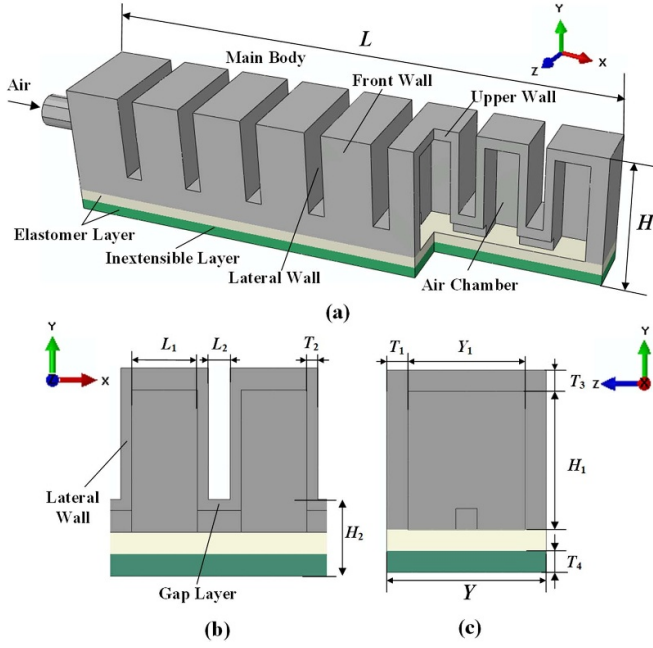


Figure 1. Schematic of the PNSA. (a) The whole structure of the PNSA. (b) Front view. (c) Side view.

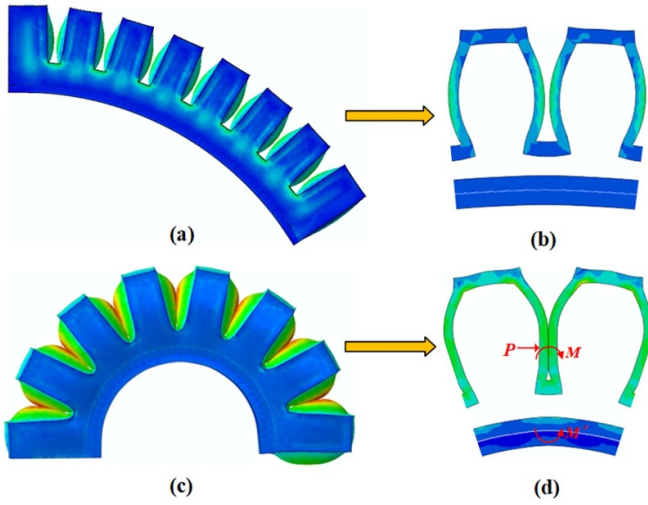


Figure 2. FE simulation results of the actuator deformation. (a) Adjacent lateral walls are not in contact. Overall structure. (b) Adjacent lateral walls are not in contact. Cross-section view. (c) Adjacent lateral walls are in contact. Overall structure. (d) Adjacent lateral walls are in contact. Cross-section view.

the length of the inextensible layer is constant, the actuator bends downward, as shown in figures 2(a) and (b). Lateral walls do not affect the bending of the actuator before adjacent lateral walls come into contact. After lateral wall contacts, lateral walls squeeze each other and generate contact force and moment, as shown in figures 2(c) and (d). The internal stress of the gap layer balances the contact force and moment, stretching the gap layer and increasing the bending angle of the actuator.

The lateral wall of the PNSA can be considered a membrane, and the contact surface of adjacent lateral walls can be

considered a rigid substrate. Since lateral walls are not parallel to the contact surface, the contact between adjacent lateral walls is essentially the contact between the inclined membrane and the substrate. The angle of adjacent lateral walls is related to the bending caused by elastomer stretching. In the next section, we first analyze the bending caused by elastomer stretching and establish a bending model.

3. Modeling of bending caused by elastomer stretching

3.1. Constitutive model

The PNSA is mostly made of rubber. In this paper, we used the Yeoh model [19] to describe actuators' mechanical properties because it has good adaptability within 100% deformation. The strain energy density function of the Yeoh model is as follows:

$$U = C_{10}(I_1 - 3) + C_{20}(I_1 - 3)^2, \quad (1)$$

where U is the strain energy function, C_{10} and C_{20} are material parameters, and I_1 is the first invariant of the three (length, width, and height) principal stretch ratios λ_1 , λ_2 , and λ_3 as

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2. \quad (2)$$

Due to the inextensible layer's propensity to bend easily but resist stretching and the relatively large aspect ratio of the actuator, the length in the width direction of the actuator is approximately unchanged. Therefore, $\lambda_2 = 1$. Considering the material is incompressible, $\lambda_1 \lambda_2 \lambda_3 = 1$, so we obtain that

$$\lambda_1 = \lambda, \quad \lambda_2 = 1, \quad \lambda_3 = \frac{1}{\lambda}. \quad (3)$$

Therefore, the strain energy density U can be expressed as

$$U = C_{10} \left(\lambda^2 + \frac{1}{\lambda^2} - 2 \right) + C_{20} \left(\lambda^2 + \frac{1}{\lambda^2} - 2 \right)^2. \quad (4)$$

The principal nominal stresses σ could be obtained as a function of U and λ_1

$$\sigma = \frac{\partial U}{\partial \lambda_1} = 2 \left(\lambda_1 - \frac{1}{\lambda_1^3} \right) \left(C_{10} + 2C_{20} \left(\lambda_1^2 + \frac{1}{\lambda_1^2} - 2 \right) \right). \quad (5)$$

3.2. Analysis of the PNSA deformation

The cross-section of the PNSA after inflation is shown in figure 3. Since our aim in this section is to analyze the bending due to elastomer stretching, we ignore the deformation of the lateral wall. The actual deformation of the actuator complex. For convenience, we propose the following simplifications and assumptions:

- Ignore pressure on the front and rear walls, upper, and bottom layers.

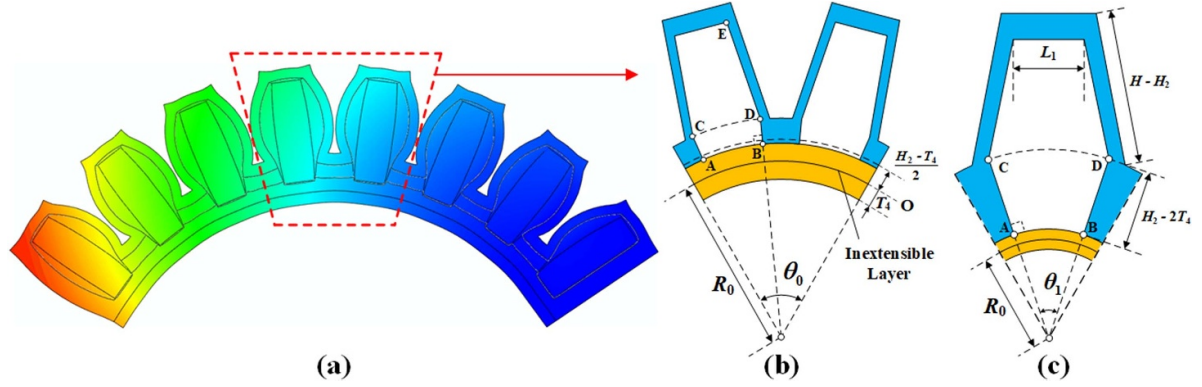


Figure 3. The cross-section of the PNSA after inflation. (a) The whole actuator. (b) Two air chambers. (c) An air chamber. For the convenience of observation, the scales of the structures in (b) and (c) differ.

- The actuator bends into a constant curvature shape.
- The gap layer is always perpendicular to the bottom surface during inflation, i.e. AC and BD are always perpendicular to the bottom surface of the air chamber.
- Most of the pressure is balanced by the stresses on the front and rear walls and the bottom layer; thus, the length of the upper wall changes very little, which is approximately assumed to be invariable and equal to L_1 .
- Since the stretching of the chamber causes structural bending, we consider that the length of the intermediate position of the gap layer is invariable.

Based on the above simplification and assumptions, we can infer that the l_{BD} and l_{DE} (we define the l_{ij} as the curve length between points i and j) are invariable during inflation, and the bending angle θ_1 of an air chamber is given by

$$\theta_1 = \frac{\theta}{N} - \frac{L_2 + 2T_2}{R_0 + 0.5(H_2 - T_4)}, \quad (6)$$

where θ is the bending angle of the actuator, N is the number of air chambers, and R_0 is the radius of curvature of the actuator, is given by

$$R_0 = \frac{L}{\theta}. \quad (7)$$

We can obtain l_{AB} and l_{CD} from figure 3:

$$l_{AB} = (R_0 + T_4)\theta_1, \quad (8)$$

$$l_{CD} = (R_0 + H_2 - T_4)\theta_1. \quad (9)$$

During inflation, the work W done by the air on the actuator is converted into the strain energy E of the elastomer, so

$$W = \int PdV_a = E = \int UdV_r, \quad (10)$$

where V_a is the volume of the air chamber and V_r is the volume of elastomer.

For silicone materials, the material parameter C_{20} is very small compared to C_{10} . Since the strain on the actuator during inflation is usually less than 100%, i.e. $1 < \lambda_1 < 2$, the negative cubic term of λ_1 in equation (5) can be ignored compared to the first-order term of λ_1 . Therefore, we can get $\sigma \approx 2C_{10}\lambda_1$ according to equation (5), which means the relationship between stress σ and strain λ_1 is approximately linear, so the work W done by the air on the actuator can be expressed by

$$W = \frac{1}{2}P\Delta V_a N, \quad (11)$$

where ΔV_a is the change in the volume of the chamber. Because the width of the chamber is considered invariable, we have

$$\Delta V_a = Y_1 \Delta S, \quad (12)$$

where ΔS is the change in the cross-sectional area of a chamber, which can be calculated from figure 3(c).

Figure 4 shows the overall structure of the actuator. We divide the PNSA into five parts with regular shapes according to the characteristics of the actuator deformation, and different colors represent different parts. The total strain energy E of the structure is equal to the sum of the strain energies of each part. We ignore the deformation of the lateral walls, and the length of the upper wall is considered invariable, so $E_E = 0$. Therefore, we can obtain that

$$E = E_A + E_B + E_C + E_D. \quad (13)$$

The strain energy can be obtained by integrating the elastomer body's infinitesimal elements in a vertical direction. From the deformed geometry of the membrane, we can obtain that

$$E_A = \int_{-T_4}^{T_4} U(\lambda_A) LY dx, \quad (14)$$

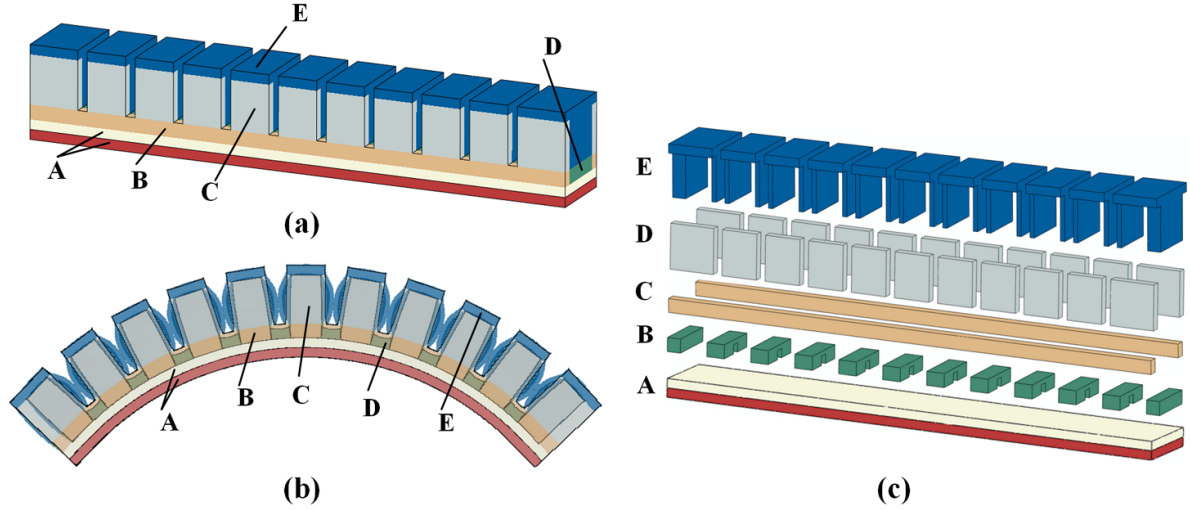


Figure 4. The PNSA is divided into five parts, and different colors represent different parts. (a) Oblique view before inflation. (b) Front view after inflation (translucent). (c) Exploded view before inflation.

$$E_B = \int_{T_4}^{H_2-T_4} 2U(\lambda_B) T_1 L dx, \quad (15)$$

$$E_C = \int_{H_2-T_4}^{H_1+T_4} 2NU(\lambda_C) T_1 (L_1 + 2T_2) dx, \quad (16)$$

$$E_D = \int_{T_4}^{H_2-T_4} NU(\lambda_D) Y_1 (L_2 + 2T_2) dx, \quad (17)$$

where λ_i is the principal stretch ratio of part i , the direction of x corresponds to the vertical direction, namely perpendicular to the bottom. The value of x represents the distance from the inextensible layer. We can obtain that

$$\lambda(x) = \begin{cases} \frac{R_0 + x}{R_0}, & -T_4 \leq x \leq H_2 - T_4 \\ kx + 1 - k(H_1 + T_4), & H_2 - T_4 \leq x \leq H_1 + T_4 \end{cases}, \quad (18)$$

where

$$k = \frac{(R_0 + H_2 - T_4)\theta_1 - L_1}{(L_1 + 2T_2)(H_2 - 2T_4 - H_1)}. \quad (19)$$

Combining the above equations, we can obtain the relationship between the bending angle θ and air pressure P when the lateral wall contact is not considered.

4. Contact modeling of the lateral wall

The contact between the lateral walls is essentially the contact between the inclined membrane and the substrate. In this part, we develop an inclined membrane contact model to analyze the contact between the lateral walls. Section 4.1 provides a detailed introduction to the inclined membrane contact model. In section 4.2, we utilize the inclined membrane contact model

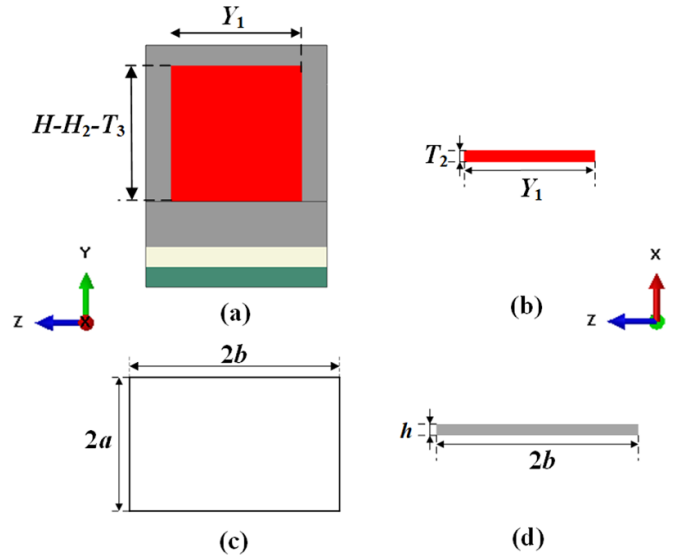


Figure 5. Schematic diagram of the lateral wall and rectangular membrane. (a) Front view of the lateral wall. (b) Top view of the lateral wall. (c) Front view of the rectangular membrane. (d) Top view of the rectangular membrane.

to determine the pressure at which adjacent lateral walls contact and analyze the deformation caused by lateral wall contact. We then integrate this model with the bending model caused by elastomer stretching to ultimately establish a comprehensive bending model for PNSAs.

4.1. Contact modeling of inclined membranes

4.1.1. Free inflation of membranes. The PNSA lateral wall can be considered a rectangular membrane. Figure 5 illustrates the schematic diagram of the lateral wall and rectangular membrane. The lengths of the lateral wall and the rectangular membrane along the Y -axis are $H - H_2 - T_3$ and $2a$, respectively. Similarly, along the Z -axis, the lengths are Y_1 for the lateral

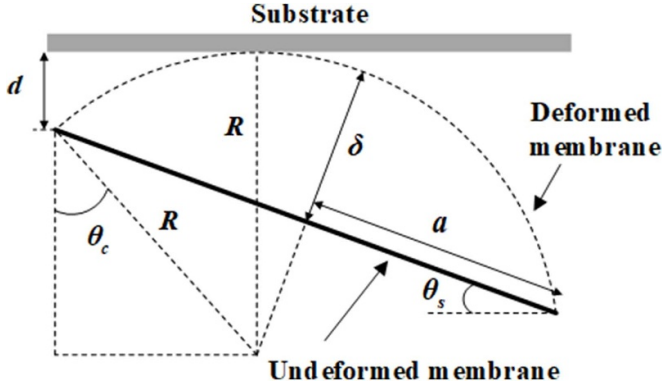


Figure 6. Deformed geometry of the flat membrane just in contact with the substrate. The labels are: $2a$ width of membrane, R radius of curvature, d distance between membrane and substrate, δ deformation deflection of the membrane, θ_c angle between the center of curvature and the vertical direction, θ_s inclination angle of the membrane.

wall and $2b$ for the rectangular membrane. The thickness of the lateral wall and rectangular membrane is T_2 and h , respectively. The boundaries of both the lateral wall and the rectangular membrane are considered fixed. Conway [20] analyzed the large deformation of the rectangular membrane. The maximum deflection of the rectangular membrane can be obtained by

$$\delta = \gamma \sqrt[3]{\frac{Pb^4}{Eh}}, \quad (20)$$

where P , γ , E , and h are pressure, a constant depending on the aspect ratio, the initial Young's modulus of the material, and the thickness of the membrane, respectively. The Young's modulus E can be expressed by

$$E = 2\mu(1 + \nu), \quad (21)$$

where μ is the shear modulus and ν is Poisson's ratio. The aspect ratio q of the PNSA's lateral wall is usually less than 2. According to [20], the relationship between γ and the aspect ratio q when $1 < q < 2$ can be expressed as

$$\gamma = 0.183q^2 - 0.9555q + 1.5905, \quad 1 \leq q \leq 2. \quad (22)$$

Substituting equation (22) into equation (20), we can calculate the maximum deflection of the membrane under pressure P .

Figure 6 is a geometric diagram of the inclined membrane with substrate. The inflated membrane is assumed to have a circular geometry. As the pressure increases, the membrane deformation gradually becomes larger. We define P_c as the pressure at which the inclined membrane just contacts with the substrate. From the deformed geometry of the membrane, we can obtain that

$$R \cos \theta_c + d = R \quad (23)$$

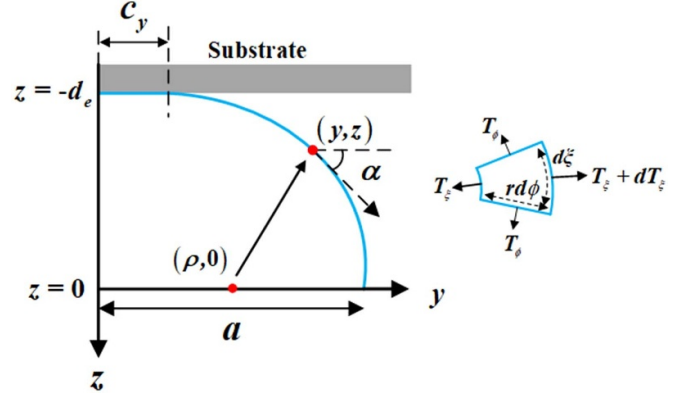


Figure 7. The geometry of a cross-section of the circular membrane before and after deformation. ξ and ϕ represent the longitudinal and latitudinal directions. c_y is the contact region's radius. The inset shows the line tensions T_ξ and T_ϕ acting on the deformed membrane.

The radius of curvature R can be expressed as

$$R = \frac{\delta^2 + a^2}{2\delta} = \frac{1}{2} \left(\delta + \frac{a^2}{\delta} \right) = \frac{a}{2} \left(\frac{\delta}{a} + \frac{a}{\delta} \right). \quad (24)$$

The θ_c can be expressed as

$$\theta_c = \frac{\pi}{2} - \theta_s - \arccos \frac{a}{R}. \quad (25)$$

By combining the above equation, we can calculate the critical pressure P_c .

4.1.2. Analysis of contact between membranes and substrates. When the pressure is greater than the critical pressure P_c , the membrane will be in contact with the substrate. The contact between inclined membranes and substrates is highly complex due to the asymmetrical deformation of the membrane. Therefore, we endeavor to transform it into an equivalent horizontal membrane–substrate contact to simplify the problem. Prior to proposing the approach of transforming inclined membrane contact into an equivalent horizontal membrane contact, we first introduce the methodology for analyzing contact between horizontal membranes and the substrate.

The geometric representation of the contact between the deformed circular membrane and the substrate is illustrated in figure 7, where the initial position of the membrane is parallel to the substrate. According to the mechanical model of the contact of the large deformation of the circular membrane [21, 22], the equations for the large deformation of the circular membrane under pressure P

$$\frac{d\lambda_\xi}{d\rho} = \frac{\lambda_\xi (T_\phi - T_\xi) \cos \alpha + \lambda_\phi (\partial T_\xi / \partial \lambda_\phi) (\lambda_\xi \cos \alpha - \lambda_\phi)}{\rho \lambda_\phi (\partial T_\xi / \partial \lambda_\xi)}, \quad (26)$$

$$\frac{d\alpha}{d\rho} = \frac{P \rho \lambda_\phi \lambda_\xi - \lambda_\xi T_\phi \sin \alpha}{\rho \lambda_\phi T_\xi}, \quad (27)$$

$$\frac{d\lambda_\phi}{d\rho} = \frac{\lambda_\xi \cos \alpha - \lambda_\phi}{\rho}, \quad (28)$$

$$\frac{dy}{d\rho} = \lambda_\xi \sin \alpha, \quad (29)$$

where λ_ξ and λ_ϕ are longitudinal and latitudinal principal stretch ratios, T_ξ and T_ϕ are longitudinal and latitudinal line tensions, and other symbols meanings refer to figure 6. The line tensions T_ξ and T_ϕ are related to the strain energy density function U

$$T_\xi = \frac{h}{\lambda_\phi} \frac{\partial U}{\partial \lambda_\xi}, \quad (30)$$

$$T_\phi = \frac{h}{\lambda_\xi} \frac{\partial U}{\partial \lambda_\phi}. \quad (31)$$

The energy density function U can usually be written as a function of λ_ξ and λ_ϕ . For example, for an incompressible NH material, the strain energy is given by

$$U = \frac{E}{6} \left(\lambda_\xi^2 + \lambda_\phi^2 + 1 / (\lambda_\xi \lambda_\phi)^2 - 3 \right). \quad (32)$$

Since the membrane in the contact region is assumed to be under uniform equi-biaxial tension, the boundary conditions for an inflated circular membrane of radius and distance from the substrate are as follows:

$$\begin{aligned} (a) \quad & y(\rho = a) = 0 \\ (b) \quad & \lambda_\phi(\rho = a) = 1 \\ (c) \quad & \alpha(\rho = \rho^*) = 0 \\ (d) \quad & y(\rho = \rho^*) = -d_e \\ (e) \quad & \lambda_\phi(\rho = \rho^*) = \lambda_\xi(\rho = \rho^*), \end{aligned} \quad (33)$$

where ρ^* is the position of the contact region boundary before deformation, and we have

$$\rho^* = \frac{c_y}{\lambda_\phi}, \quad (34)$$

where c_y is the contact area radius.

By applying specified boundary conditions, we numerically integrate solutions for equations (26)–(29) using an ordinary differential equation solver. This integration helps determine the radius (c_y) of the contact region between the horizontal circular membrane and the substrate, as well as its location. Notably, the equation can be solved by specifying either ρ^* with a pressure to be determined or the pressure with a value of ρ^* to be determined.

While equations (26)–(29) precisely govern the contact between the horizontal membrane and substrate, the complex mechanical analysis involved in inclined membrane deformation requires certain assumptions and simplifications in our subsequent analysis. This transforms the problem into a contact between a horizontal membrane and the substrate.

Figure 8 illustrates the contact deformations between inclined and horizontal membranes and substrates. Figures 8(a) and (b) present FE simulation results, while

figures 8(c) and (d) provide schematic diagrams. The transparent membranes in figures 8(a) and (b) and the dashed lines in figures 8(c) and (d) represent freely inflated membranes at the same pressure. The geometric and material parameters of the inclined and horizontal membranes are identical, and the pressure on both membranes is the same. Therefore, the freely inflated inclined membrane mirrors the freely inflated horizontal membrane.

The system's state remains independent of the load application order, allowing us to consider the substrate as fixed. The membrane inflation can be envisioned as the result of the substrate gradually moving to its actual position from a distance while the deformed membrane inflates freely. Since the freely inflated inclined membrane and horizontal membrane in figure 8 are identical, we conjecture that the contact area of the inclined membrane closely matches the contact area of the horizontal membrane when the substrate is squeezed downward by the same distance. In our subsequent work, we aim to validate this conjecture.

For a rectangular membrane, the contact region is assumed to be elliptical, with the lengths along the Y -axis and Z -axis defined as $2c_y$ and $2c_z$. The deformed rectangular membrane can be considered to have been obtained by stretching the deformed square membrane with the same deflection in the length direction. Therefore, we only analyze the deformation of the square membrane. For a square membrane, the contact region is assumed to be circular. We define the non-contact region on the membrane as the free surface. The equilibrium equation for the free surface [23] is as follows:

$$\kappa_\xi T_\xi + \kappa_\phi T_\phi = P, \quad (35)$$

where κ_ξ and κ_ϕ are longitudinal and latitudinal curvatures, and T_ϕ and T_ξ are longitudinal and latitudinal line tensions.

Consider the contact between membrane and substrate to be frictionless and adhesionless, in which case the membrane in the contact region is under uniform equi-biaxial tension, thus the longitudinal and latitudinal tensions at the contact edge are the same. Points A and B, located near the contact edge of the inclined membrane and substrate (refer to figure 8(c)), exhibit the same line tension at the contact edge due to the continuity of membrane tension [21], so we have

$$T_{\xi A} = T_{\phi A} = T_{\xi B} = T_{\phi B} = T_b, \quad (36)$$

where T_b is the line tension at the contact edge.

The latitudinal curvature of points A and B is equal to the latitudinal curvature at the contact edge, so we have

$$\kappa_{\phi A} = \kappa_{\phi B} = \frac{1}{c_y}. \quad (37)$$

Substituting equations (36) and (37) into equation (35), we have

$$\kappa_{\xi A} = \kappa_{\xi B} = \frac{P - T_b/c_y}{T_b}. \quad (38)$$

Equation (38) indicates that the longitudinal curvature of the surface near the contact edge is the same. Since the free

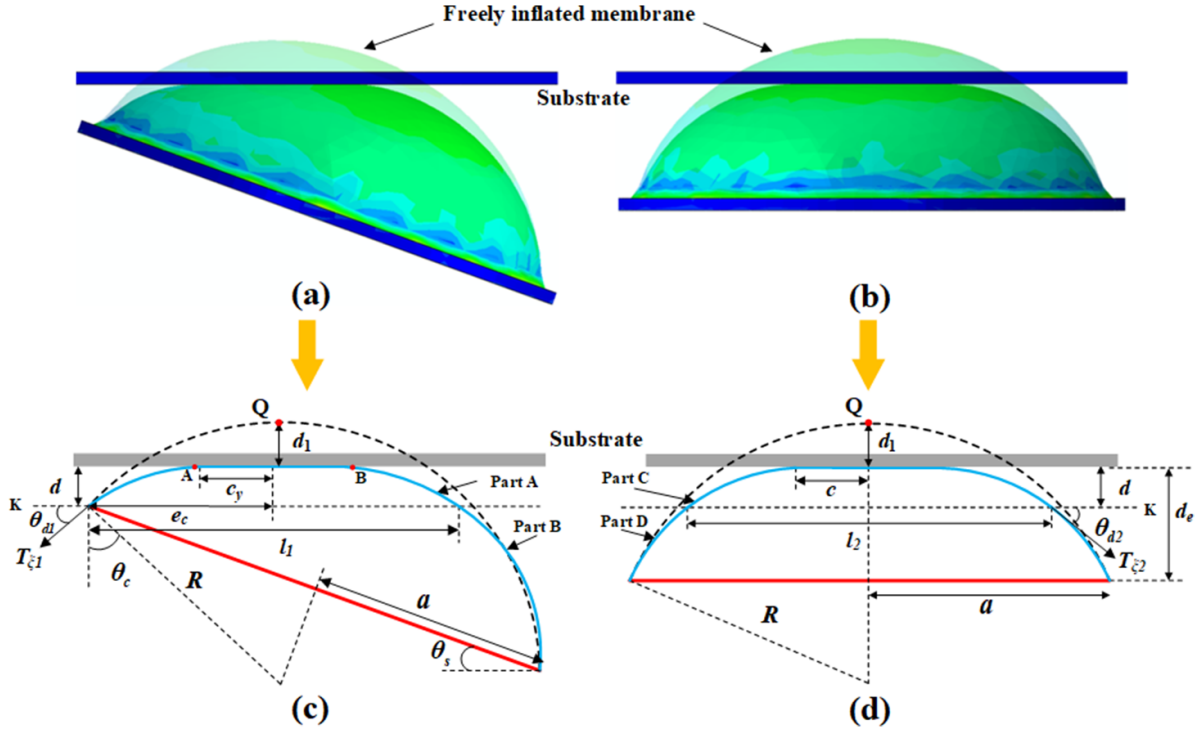


Figure 8. Contact deformations between inclined and horizontal membranes and the substrate. (a) FE simulation result of contact between inclined membrane and substrate. (b) FE simulation result of contact between horizontal membrane and substrate. (c) Schematic diagram of contact between inclined membrane and substrate. (d) Schematic diagram of contact between horizontal membrane and substrate. The transparent membranes in (a) and (b) are freely inflated membranes at the same pressure. The solid red lines in (c) and (d) represent undeformed membranes, the solid blue lines represent actual deformed membranes, and the dashed lines represent the freely inflated membranes at the same pressure. Q is the tangent point of a line parallel to the substrate and the freely inflated membrane.

surface is assumed to have a circular geometry, the longitudinal curvature of the free surface is the same. Since the contact angle between the free space and the contact region is zero [21], the shape of the deformed membrane is unique when the shape of the contact region is determined.

The point Q is the tangent point of a line parallel to the substrate of the freely inflated membrane, and the distance of Q from the substrate of the inclined membrane and the substrate of the horizontal membrane is the same. From the deformation geometry, we can obtain the distance d_e between the horizontal membrane and the substrate:

$$d_e = R \cos \theta_c + d - \sqrt{R^2 - a^2}. \quad (39)$$

The line K in figures 8(c) and (d) are lines at a distance d from the substrate, which divides the deformed inclined membrane into two parts A and B and the deformed horizontal membrane into two parts C and D. The l_1 is the distance between the two intersection points of line K and the deformed inclined membrane, and the l_2 is the distance between the two intersection points of line K and the deformed horizontal membrane. We let the contact region radius between the horizontal membrane and substrate be c . From the deformation geometry, we can obtain:

$$l_2 = 2c + 2\sqrt{\frac{d(a-c)^2}{d_e} + dd_e - d^2}. \quad (40)$$

We let the length of the cross-sectional curve of the deformed horizontal membrane (solid blue line in figure 8(d)) be l_b . The expression for l_b is related to the value of d_e , when $d_e \leq a - c$

$$l_b = \frac{(a-c)^2 + d_e^2}{d_e} \operatorname{arccot} \left(\frac{a-c}{2d_e} - \frac{d_e}{2a-2c} \right) + 2c, \quad (41)$$

when $d_e > a - c$

$$l_b = \frac{(a-c)^2 + d_e^2}{d_e} \left(\pi - \arcsin \frac{2(a-c)d_e}{(a-c)^2 + d_e^2} \right) + 2c. \quad (42)$$

The contact region radius of the inclined membrane is unknown; we assume it is equal to the contact region radius of the horizontal membrane, i.e. $c_y = c$. From the deformation geometry, we can obtain that

$$\begin{aligned} (2a \cos \theta_s - c - 0.5l_1)^2 + \left(0.5d + 2a \sin \theta_s - \frac{(0.5l_1 - c)^2}{2d} \right)^2 \\ = \left(0.5d + \frac{(0.5l_1 - c)^2}{2d} \right)^2. \end{aligned} \quad (43)$$

We let the length of the cross-sectional curve of the deformed inclined membrane (solid blue line in figure 8(c)) be l_a , which can be expressed by

Table 1. Several cases of numerical solutions for equations.

	a (mm)	d (mm)	θ_s (rad)	d_e (mm)	c (mm)	l_1 (mm)	l_2 (mm)	l_a (mm)	l_b (mm)	l_1/l_2	l_a/l_b
Case 1	10	1	0.3	4.2	4.7	16.6	16.2	23.8	23.3	1.025	1.021
Case 2	15	2	0.4	7.4	7.1	25.5	24.7	39.4	38	1.032	1.037
Case 3	20	1	0.2	4.9	15	36.8	36	46.3	45.5	1.022	1.018
Case 4	25	3	0.3	10.3	17.1	48.4	46.9	66.7	64.2	1.032	1.039

$$l_a = 2c + \frac{(0.5l_1 - c)^2 + d^2}{2d} \left(\operatorname{arccot} \left(\frac{l_1 - 2c}{4d} - \frac{d}{l_1 - 2c} \right) + \pi \right) - \arcsin \frac{2d(2a \cos \theta_s - c - 0.5l_1)}{(0.5l_1 - c)^2 + d^2}. \quad (44)$$

Obtaining symbolic solutions for the equations is quite challenging; therefore, we obtained numerical solutions by substituting some values to compare l_1 , l_2 , l_a and l_b . Table 1 presents several cases of numerical solutions, where d_e and c are the corresponding values obtained by substituting geometric and material parameters into our previously established model, and l_1 , l_2 , l_a and l_b are derived by solving equations (40)–(44). The results indicate that the l_1 and l_2 have very small differences, and the l_a and l_b also exhibit small discrepancies. Therefore, in this study, we assume that l_1 is equal to l_2 and l_a is equal to l_b . Consequently, part A and part C can be considered to have the same shape, and the lengths of the horizontal membrane and the inclined membrane cross-sectional curve are the same, and thus $\theta_{d1} = \theta_{d2}$. The membrane is assumed to be uniformly stretched, and the line tension can be approximated as related to the length of the cross-section curve, so we have

$$T_{\xi 1} = T_{\xi 2} \quad (45)$$

Since the contact angle between the free space and the contact region is zero, the combined force of the free space in the contact region is zero. Analyzing the force of part C, which is in equilibrium with forces in the vertical direction, we have

$$P_1 \pi \left(\frac{l_1^2}{4} - c^2 \right) = \pi l_2 T_{\xi 2} \sin \theta_{d2}, \quad (46)$$

where P_1 is the pressure on the inclined membrane and the horizontal membrane.

Since $l_1 = l_2$, $\theta_{d1} = \theta_{d2}$, and $T_{\xi 1} = T_{\xi 2}$, we can obtain that

$$P_1 \pi \left(\frac{l_1^2}{4} - c^2 \right) = \pi l_1 T_{\xi 1} \sin \theta_{d1}. \quad (47)$$

Equation (47) indicates that the forces are in the vertical direction of part A are balanced when the contact region radius of the inclined membrane is c .

Assume that the contact region radius c_y of the inclined membrane is greater than c , which is c_1 . Increase the pressure on the horizontal membrane until its contact region radius is equal to c_1 , corresponding to the pressure of P_2 ($P_2 > P_1$), and the l_2 , θ_{d2} , $T_{\xi 2}$ and l_b changed to l'_2 , θ'_{d2} , $T'_{\xi 2}$ and l'_b .

Analyzing the force of part C, the part C is in equilibrium with forces in the vertical direction, so we have

$$P_2 \pi \left(\frac{l'^2_2}{4} - c^2_1 \right) = \pi l'_2 T'^2_{\xi 2} \sin \theta'_{d2}. \quad (48)$$

Since the contact region radius of the inclined membrane and the horizontal membrane is both c_1 , it can be considered that l'_2 is approximately equal to l_1 , and l'_b is approximately equal to l_a . Therefore, part A and part C can be considered to have the same shape, and the lengths of the horizontal membrane and the inclined membrane cross-sectional curve are the same, and thus $\theta_{d1} = \theta'_{d2}$ and $T_{\xi 1} = T'_{\xi 2}$, so we can obtain that

$$P_1 \pi \left(\frac{l^2_1}{4} - c^2_1 \right) < P_2 \pi \left(\frac{l'^2_2}{4} - c^2_1 \right) = \pi l_1 T_{\xi 1} \sin \theta_{d1}. \quad (49)$$

Equation (49) indicates that the forces in the vertical direction of part A are not balanced when $c_y > c$, so $c_y \leq c$. Similarly, the contact region radius of the inclined membrane cannot be less than c . Because the forces in the vertical direction of part A are balanced when the contact region radius of the inclined membrane is c , the contact region radius can only be equal to c , which means the contact area of the inclined membrane is the same as the contact area of the horizontal membrane when the distance between the horizontal membrane and the substrate is d_e . Therefore, we can determine the contact area of the inclined membrane by calculating the contact area of the horizontal membrane. However, equations (26)–(29) cannot be directly applied to calculate the contact area of the inclined membrane, as equations (26)–(29) pertain to circular membranes, while our study focuses on rectangular membranes with larger deformations under the same pressure. Therefore, P in equation (27) should be replaced by the equivalent pressure P_e corresponding to a circular membrane with the same deflection as the rectangular membrane. For a circular membrane [21], the deflection δ

can be obtained by

$$\frac{Pa}{Eh} = \frac{4\delta/a}{3 \left(1 + (\delta/a)^2 \right)} \left(1 - \frac{1}{\left(1 + (\delta/a)^2 \right)^3} \right). \quad (50)$$

Substituting equation (20) into equation (50), we can derive the equivalent pressure P_e , and subsequently determine the contact region width $2c_y$ between the inclined membrane and the substrate. The Z-axis contact region length $2c_z$ can be obtained by

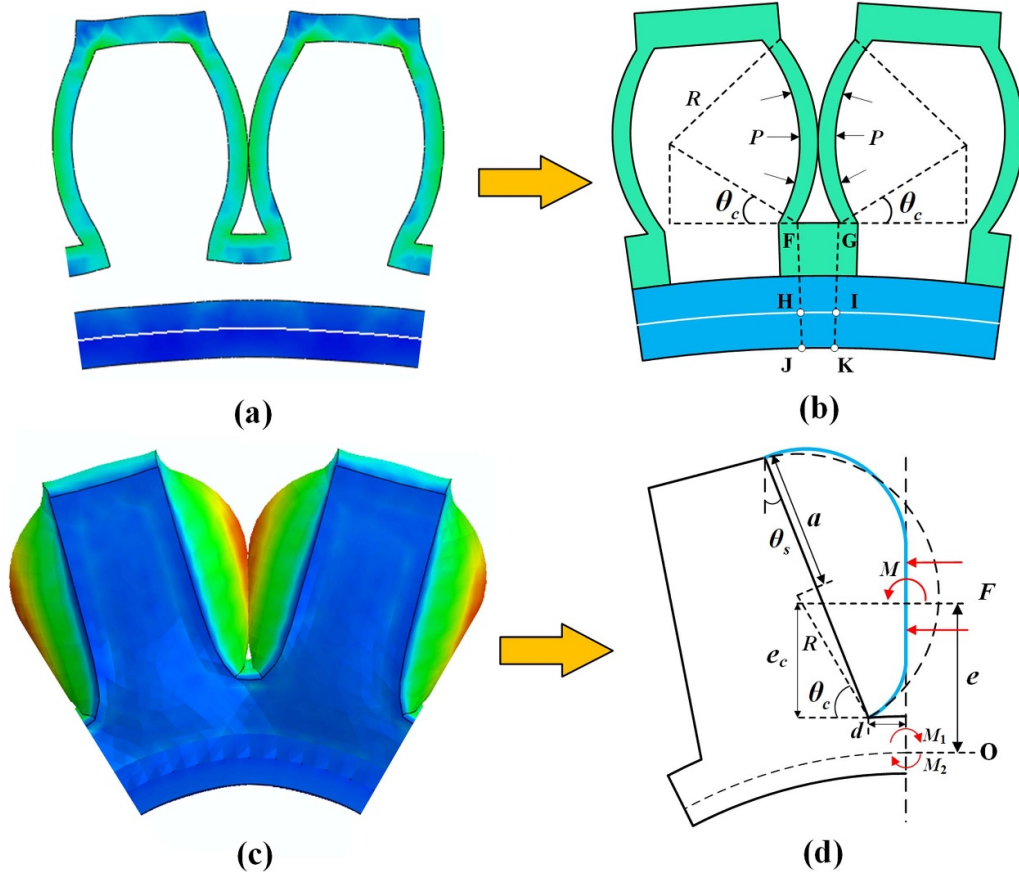


Figure 9. Schematic diagram of deformed lateral walls. (a) FE simulation cross-section when lateral walls are in exact contact. (b) Geometric analysis diagram when lateral walls are in exact contact. (c) FE simulation results when lateral walls are in contact. (d) Schematic diagram of deformed lateral walls in contact. The solid blue line is the actual deformed lateral wall, and the dashed line is the freely inflated lateral wall.

$$c_z = \frac{b}{a} c_y. \quad (51) \quad \text{middle regarded as the substrate. From figure 5, we can obtain that}$$

The contact region area S can be written as

$$S = \pi c_y c_z. \quad (52)$$

The contact force F can be written by

$$F = PS. \quad (53)$$

The e_c is distance from the center of the contact region to the interior of the inclined membrane, from figure 8(c), we can obtain that

$$e_c = \frac{l_1}{2} = c_y + \sqrt{\frac{d(a - c_y)^2}{d_e} + dd_e - d^2}. \quad (54)$$

$$a = \frac{H - H_2 - T_3}{2}, \quad (55)$$

$$b = \frac{Y_1}{2} \quad (56)$$

According to the geometric relationship obtained in section 3.2, we can obtain

$$d = \frac{l_{FG}}{2} = \frac{L_2(L + (H_2 - T_4)\theta)}{2(L + T_4\theta)}, \quad (57)$$

$$\theta_c = \arccos \frac{(R_0 + H_2 - T_4) \sin \theta_1 - 0.5L_1}{H - H_2 - T_3} - \frac{\theta}{2N} - \arccos \frac{H - H_2 - T_3}{2R}. \quad (58)$$

4.2. Modeling of the lateral wall contact

4.2.1. Calculation of critical contact pressure. Figures 9(a) and (b) are schematic diagrams where the adjacent lateral walls are in exact contact. We consider the lateral walls as rectangular membranes, with the contacting surface in the

By substituting equations (55)–(58) into the previously derived equation, we can obtain the critical pressure P_c . It should be noted that the critical pressure may not exist, i.e. the equation has no solution, which means that the lateral walls cannot come into contact regardless of the applied pressure.

4.2.2. Analysis of adjacent lateral wall contact. Figures 9(c) and (d) are schematic diagrams of the contact between adjacent lateral walls. Because the structure is symmetrical, there is only one lateral wall in figure 9(d). From the deformed geometry, we can get

$$d = \left(\frac{L_2}{\theta_g} + H_2 - T_4 \right) \sin \frac{\theta_g}{2}, \quad (59)$$

$$\theta_s = \arccos \frac{(R + H_2 - T_4) \sin \theta_1 - 0.5L_1}{H - H_2 - T_3} - \frac{\theta_1 + \theta_g}{2}, \quad (60)$$

where θ_g is the bending angle of the gap layer. The length and width of the lateral wall are invariant and can be determined by equations (55) and (56).

By substituting the above equations into the inclined membrane contact model, we can obtain the contact force F between adjacent lateral walls.

The bending moment M induced on the inextensible layer by force F is

$$M = Fe, \quad (61)$$

where e is the eccentricity of the axial force F , it can be expressed by

$$e = e_c + H_2 - T_4. \quad (62)$$

The bending moment M is balanced by the bending moment generated by the internal stress of the gap layer, so

$$M = M_1 + M_2, \quad (63)$$

where M_1 is the bending moment generated by the elastomer stretching and M_2 is the bending moment generated by the elastomer compressing because the length of the inextensible layer is invariable. The M_1 and M_2 can be calculated as

$$M_1 = \int_0^{H_2-T_4} \sigma Y x dx, \quad (64)$$

$$M_2 = \int_{-T_4}^0 \sigma Y x dx, \quad (65)$$

where internal stress σ can be obtained by substituting the gap layer's principal stretch ratio λ_g into equation (5). The λ_g is given by

$$\lambda_g = \frac{L_2 + x\theta_g}{L_2 + x\theta_{FG}}, -T_4 \leq x \leq H_2 - T_4, \quad (66)$$

where θ_{FG} is the bending angle of the gap layer caused by elastomer stretching and is given by

$$\theta_{FG} = \frac{L_2\theta}{L + T_4\theta}. \quad (67)$$

The effect of the lateral wall extrusion on the air chamber can be neglected, so we can obtain the bending angle of the actuator under air pressure P

$$\theta' = \begin{cases} f(P), & P \leq P_c \\ f(P) + (n-1)(\theta_g - \theta_{FG}), & P > P_c \end{cases}, \quad (68)$$

where $f(P)$ is the bending angle of the actuator caused by elastomer stretching.

5. Validation of the theoretical model

In this part, the theoretical model is verified through FEA and experiments. The models that we validated include the bending model due to elastomer stretching, the critical contact pressure model, the inclined membrane contact model, and the bending model considering lateral wall contact. In section 5.1, we introduce the details of simulations and experiments. In section 5.2, we present and discuss the comparison results between the theoretical model and other methods.

5.1. Simulations and experiments

5.1.1. FE simulations. We carried out simulation analysis using FE simulation software; details regarding the process of the FE simulation can be found in [24]. We use ten-node quadratic tetrahedral elements (C3D10) for meshing, and the global mesh seed is set to 1 mm. Gravity is not considered in the model. Silicone was set as a hyperelastic material, and the Yeoh model was used to describe the mechanical behavior of silicone, with the parameters $C_{10} = 0.116$ mPa and $C_{20} = 0.025$ mPa, consistent with the material used in the experiment. The Young's modulus E of the membrane was set to 0.696 mPa and the Poisson's ratio to 0.499. The Young's modulus of the inextensible layer was set to 6.5 GPa and the Poisson's ratio to 0.2. We verified the bending model, the critical pressure, and the inclined membrane contact model using FEM. The FE simulation results of the actuator at different air pressures are shown in figure 10(b). The parameters of the actuator are listed in table 2. Figure 11(b) shows the FE simulation result of the inclined membrane in contact with the substrate. The parameters of the inclined membrane are shown in table 3, which can represent the general lateral walls of PNSAs.

5.1.2. Experiments. We used silicon rubber E630# with a Shore hardness of 30° degrees to fabricate membranes and actuators to conduct experiments. We conducted uniaxial tensile tests on rubber according to the ASTM D412 standard using the MTS universal testing machine to determine the material parameters of rubber. We measured the Young's modulus $E = 0.696$ mPa, $C_{10} = 0.116$ mPa, and $C_{20} = 0.025$ mPa.

We designed an experimental device to conduct contact experiments between inclined membranes and substrates. In order to facilitate the observation of the contact region, we

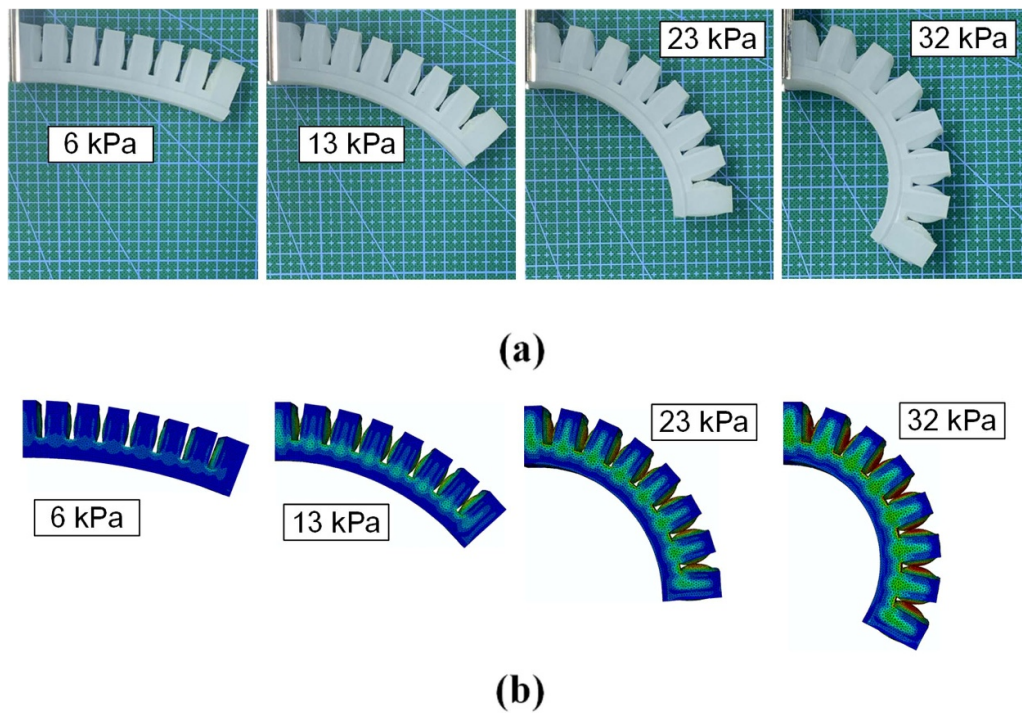


Figure 10. The PNSA bends under different air pressures. (a) Experimental. (b) FE simulation.

Table 2. Parameters of the Actuator.

Parameter	Value	Parameter	Value
L (mm)	96	T_1 (mm)	2.5
Y (mm)	18	T_2 (mm)	1.5
H (mm)	24	T_3 (mm)	2
L_1 (mm)	6	T_4 (mm)	2
L_2 (mm)	3	C_{10} (mPa)	0.116
Y_1 (mm)	13	C_{20} (mPa)	0.025
H_1 (mm)	18	N	8
H_2 (mm)	8.5		

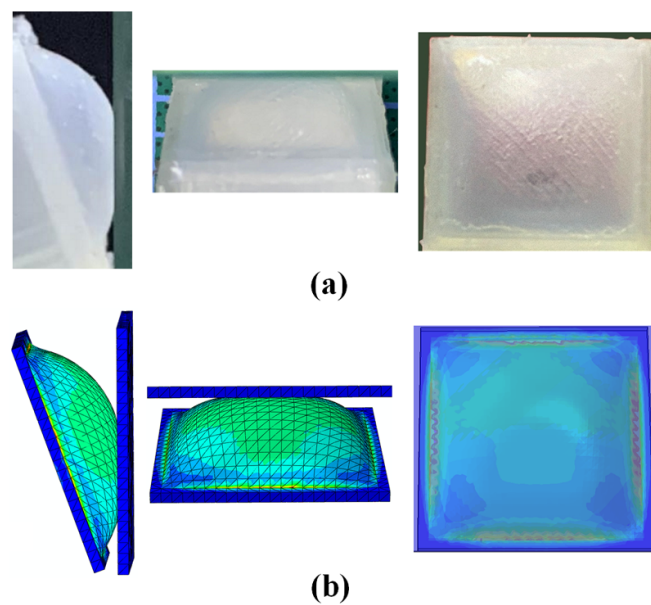


Figure 11. Contact between the inclined membrane and the substrate. From left to right: front view, top view, side view. (a) Experimental. (b) FE simulation.

Table 3. Parameters of the inclined membrane.

Parameter	Value	Parameter	Value	Parameter	Value
a (mm)	10	d (mm)	1.5	h (mm)	1
θ_s (°)	20	E (mPa)	0.696	P (kPa)	25

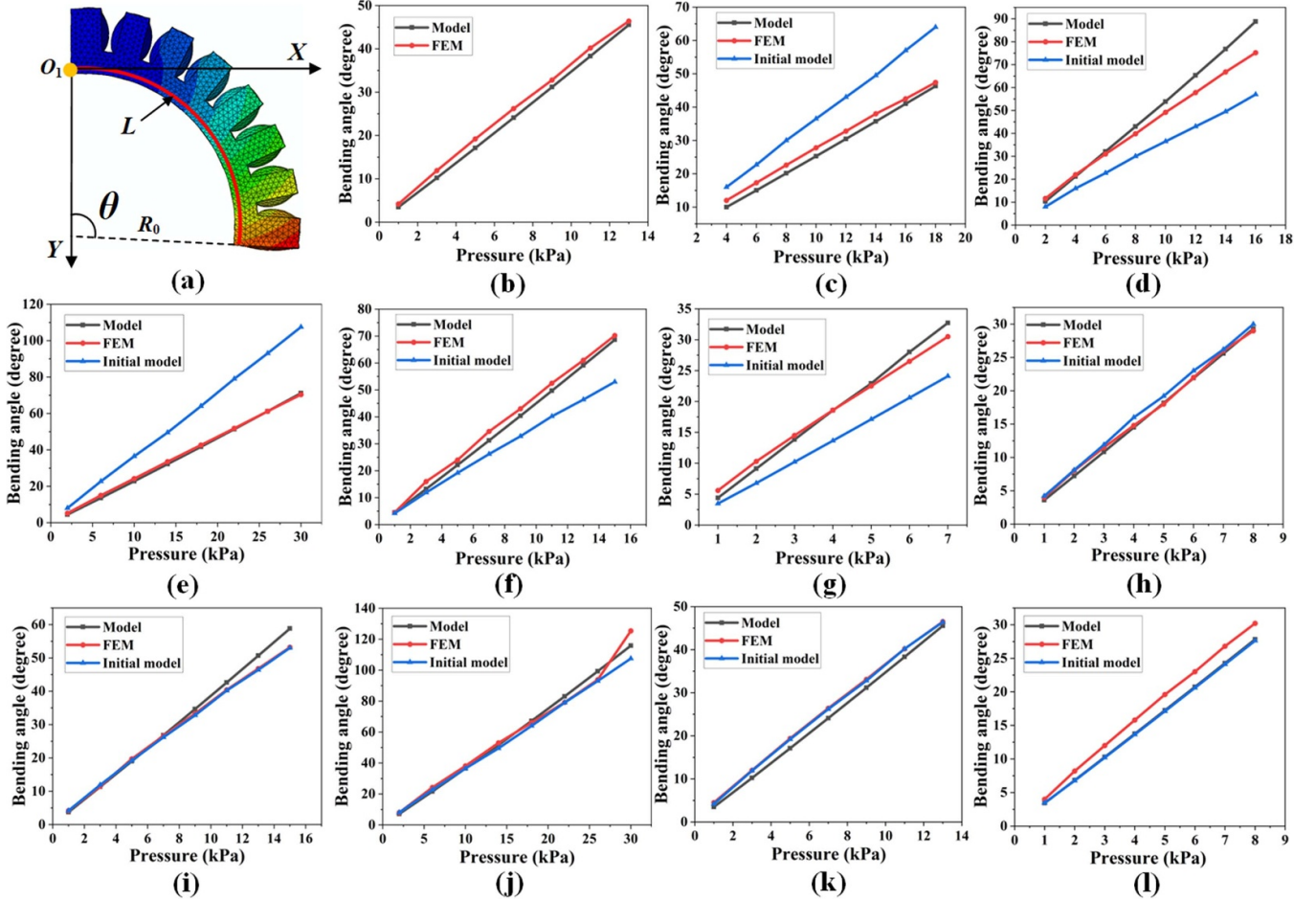


Figure 12. Comparison results of the bending model caused by elastomer stretching and FEM. (a) Deformation result of FE simulation. (b) Initial PNSA. (c) $C_{10} + 0.04$ mPa. (d) $T_1 - 1$ mm. (e) $Y_1 - 3$ mm. (f) $T_4 - 2$ mm. (g) $H_2 - 2$ mm. (h) $T_2 + 0.5$ mm. (i) $L_1 + 1$ mm. (j) $L_2 + 1$ mm. (k) $T_3 + 1$ mm. (l) $H_1 + 2$ mm. The blue curves in (c)–(l) represent the bending angle of the initial actuator.

used transparent glass as a substrate. The experimental snapshots of the inclined membrane in contact with the substrate are illustrated in figure 11(a). Additionally, we fabricated PNSAs with different parameters and conducted a series of experiments. Figure 10(a) shows the experimental results of the actuator under different air pressures, and the parameters of the actuator are shown in table 2. The bending plane of the actuator is parallel to the ground, so the effect of gravity on the actuator can be neglected.

5.2. Results and discussions

5.2.1. Bending model caused by elastomer stretching. We modified all the parameters of the actuator to validate our bending model caused by elastomer stretching, including ten

geometric parameters and one material parameter. Due to the large number of parameters, experiments require the production of a large number of actuators, which is too expensive. Because FEM is very accurate, we only compared our bending model with FEM. Figure 12 shows the comparison results for the bending angle before the lateral walls contact to eliminate the influence of the lateral walls on bending. Figure 12(a) shows the comparison result of the bending angle of the initial actuator; the parameters of the initial actuator are shown in table 2. It can be seen that our model is very close to FEM. Figures 12(c)–(l) show comparison results of the actuator changing parameters, where the blue curve represents the initial actuator to compare. The results show that our model has strong robustness for each parameter of the actuator, and changing T_1 , T_4 , Y_1 , C_{10} and H_2 can significantly change

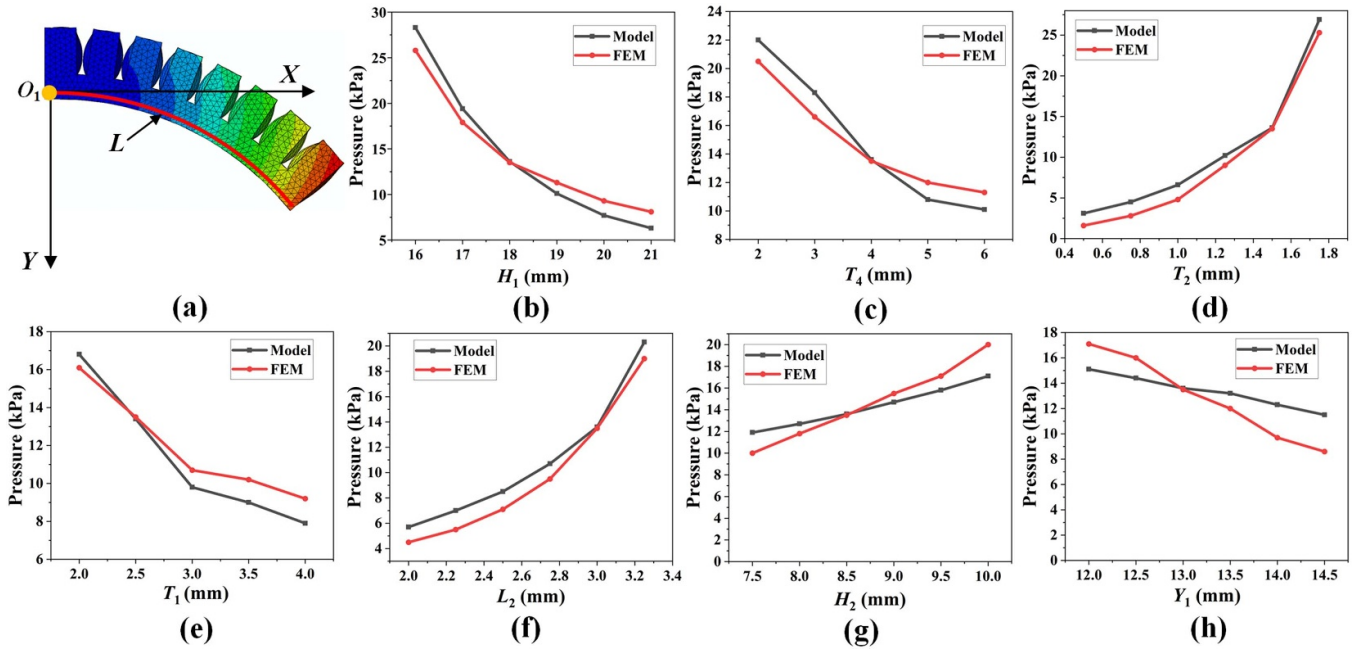


Figure 13. Comparison results of critical air pressure obtained from the theoretical model and the FEM. (a) The result of the FE simulation when lateral walls are exact in contact. (b) H_1 . (c) T_4 . (d) T_2 . (e) T_1 . (f) L_2 . (g) H_2 . (h) Y_1 .

the bending angle, while changing other parameters has little effect.

5.2.2. Critical contact pressure. Figure 13 illustrates the comparison results for critical pressure between our model and FEM, where the horizontal axis is the value of the changed parameter and the vertical axis is critical contact pressure. We changed seven geometric parameters of PNSAs instead of all parameters to validate the model, as other parameters have little impact on lateral wall deformation and structural bending or are easy to analyze (material parameter C_{10}). The results show our model can accurately predict critical pressures. For parameters H_1 , T_1 , T_2 , T_4 , and L_2 , our model is very close to the FEM results, but for Y_1 and H_2 , there are some errors between the theoretical model and the FEM.

5.2.3. Inclined membrane contact model. We compared our inclined membrane contact model with FEM and experimentation. Furthermore, given that Sachin *et al* [18] have also proposed a contact model for inclined membranes with the substrate, we have compared our model with theirs. The results of Sachin *et al*'s model were calculated based on the method provided in [18]. We compared the following six physical quantities: deflection of free expansion δ , critical air pressure P_c , length of contact region in vertical direction c_y , length of contact region in horizontal direction c_x , area of contact region S , eccentricity distance e_c .

The parameters of inclined membranes are shown in table 3. Figure 14 illustrates the comparison results, where the horizontal axis represents the aspect ratio q ranging from 1.0 to 1.6. The results show that our model can accurately predict each physical quantity and exhibits higher accuracy compared

to Sachin *et al*'s model. The physical quantities that impact PNSA bending are the contact area S and the eccentricity distance e_c . It can be seen that we can predict S accurately when q is 1. Although the error increases as the aspect ratio increases, the maximum error is less than 30%. For e_c , the maximum error of our model is less than 10%.

5.2.4. Bending model considering lateral wall contact.

Figure 15 shows the comparison results of four different PNSAs, where model 1 represents the bending model caused by elastomer stretching and model 2 represents the bending model considering the lateral wall contact. The red region on the left of figure 15 represents before lateral wall contact, and the green region on the right represents after lateral wall contact. It can be seen that the four curves are very close before the lateral wall contact. However, when pressure exceeds the critical contact pressure, the bending model caused by elastomer stretching yields noticeably smaller results, which can be attributed to the influence of the lateral wall contact. Our theoretical bending model is very close to FEM and experimental results, which can effectively predict the bending angle of the PNSA.

5.2.5. Error analysis and discussions. Table 4 shows the average percentage error between the theoretical model, and the FE simulation, and the experimental results. In table 4, the 'Error' of model-1 and model-2 represents the error between the model and FE simulation, the 'Error-1' of model-3 and model-4 represents the error between the model and the experiment, and the 'Error-2' represents the error between FE simulation and the experiment. Type-a, b, c, and d represent the four types of drivers used in model 4 experiments.

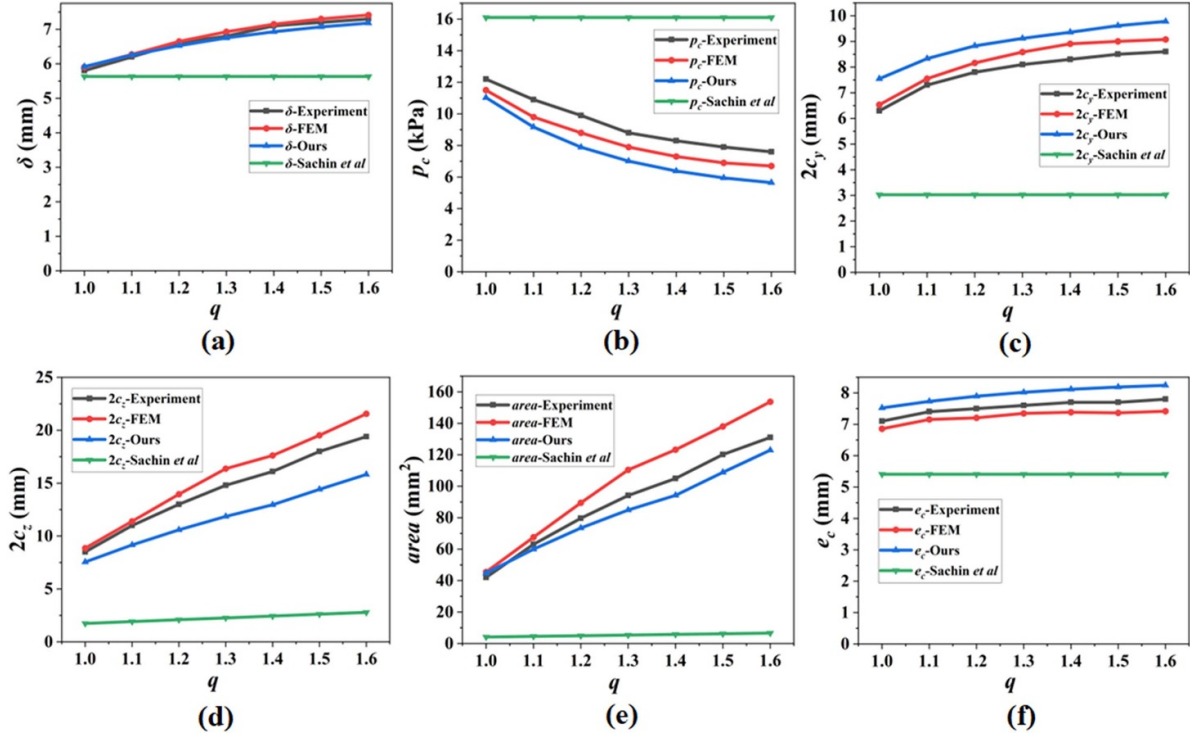


Figure 14. Comparison results for inclined membrane contact. (a) δ . (b) P_c . (c) $2c_y$. (d) $2c_z$. (e) S . (f) e_c .

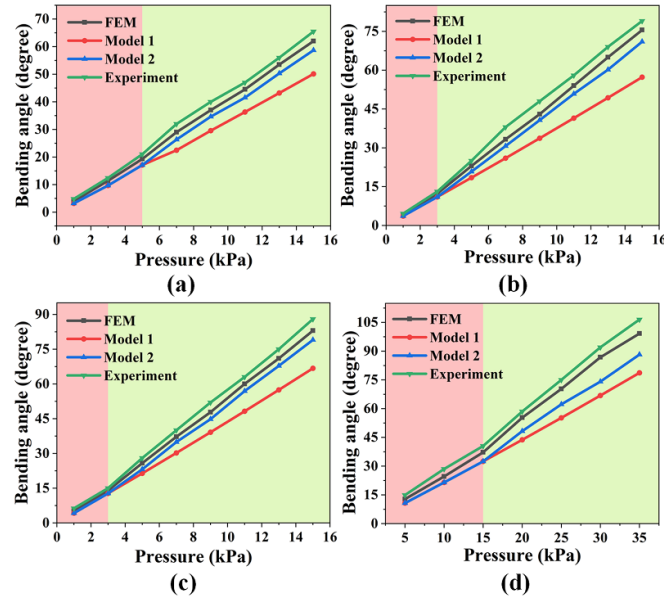


Figure 15. Comparison results of the bending model without considering lateral wall contact (model 1), the bending model considering lateral wall contact (model 2), FEM, and experiment. (a) $L_2 = 1$ mm. (b) $L_1 = 1.6$ mm, $T_2 = 0.8$ mm. (c) $L_1 = 1.6$ mm, $T_1 = 0.5$ mm, $T_2 = 0.8$ mm. (d) $T_1 = 1.5$ mm.

The overall accuracy of our model is relatively high, demonstrating effective prediction of actuator deformation. The bending model induced by the stretching of the elastic material is quite accurate, with an overall error of approximately 6%. However, there is some error in predicting the critical pressure, possibly related to distortions at the edge of the

elastic material. The overall error of the inclined membrane contact model is about 10%, with significant variations in prediction accuracy for different parameters. For the prediction of the overall bending angle of the driver, our model has a prediction accuracy of about 15%–20%, and the FEM accuracy is about 8%.

Table 4. The average percentage error between theoretical models, finite element simulations, and experiments.

		$C_{10} + 0.04$ (mPa)	$T_1 - 1$ (mm)	$Y_1 - 3$ (mm)	$T_4 - 2$ (mm)	$H_2 - 2$ (mm)	$T_2 + 0.5$ (mm)	$L_1 + 1$ (mm)	$L_2 + 1$ (mm)	$T_3 + 1$ (mm)	$H_1 + 2$ (mm)
Model-1	Initial										
Error		8.8%	8.7%	5.4%	4.3%	6.5%	1.8%	3.5%	1.7%	6.6%	10.1%
Model-2	H_1	T_4	T_2	T_1	L_2	H_2	Y_1				
Error		11.5%	7.8%	25.6%	7.9%	15.7%	9.1%	15.5%			
Model-3	δ	P_c	c_y	c_z	S	e_c					
Error-1		1.5%	20.0%	17.7%	5.5%	7.8%	16.9%				
Error-2		1.3%	10.55%	7.8%	3.9%	13.5%	13.5%				
Model-4	Type-a	Type-b	Type-c	Type-d							
Error-1		17.2%	15.4%	15.2%	20.5%						
Error-2		8.1%	7.8%	7.9%	8.6%						

6. Conclusion

This paper presents a flexible theoretical framework for modeling the mechanics of PNSAs. The key innovation was an inclined membrane contact model that transformed complex 3D interactions into equivalent planar contacts. This analytical approach enabled the prediction of coupled deformation modes under large strains.

The integrated model provides fundamental insights into relationships between actuator geometry, material properties, and deformation kinematics. By bridging fundamental principles from elasticity and contact theories, this work enhances understanding of soft machine mechanics. The methodology establishes a generalized modeling tool applicable to diverse pressure-driven soft robots.

Additionally, the novel contact model has significant potential utility for analyzing multidimensional contact problems across disciplines. The technique introduced can be extended to study the interfacing and friction of compliant mechanisms and surfaces across manufacturing, materials, and mechanics. This could enable the characterization of complex phenomena like adhesion and surface damage accumulation.

In summary, the theoretical developments presented make fundamental contributions to the science of soft machines and contact mechanics. The flexible modeling approach provides an analytical foundation for the design, control, and performance optimization of pneumatic soft robots. It also gives a framework to build upon in tackling open questions of contact interactions central to machine theory. Nevertheless, the inclined membrane contact model proposed in this paper still has its limitations. The essence of the model is transformed for analysis into a horizontal membrane, involving some simplifications and assumptions in the modeling process and lacking rigorous mechanical proof. Additionally, the actual deformation shape of the rectangular membrane is highly complex, and we approximate it as an ellipse, which is also a drawback of the model. The model's error is approximately 10%, indicating room for improvement. Future work includes establishing a rigorous mechanical model for inclined membrane contact to enhance accuracy.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Conflict of interest

The authors declare no conflict of interest.

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